SignSync

Bridging Voices, On-Device

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Team **sazi**

iQOO Hackathon 2026

On-device AI

No cloud.
No upload.
No compromise.

2.78 Million Voices, Zero On-Device Tools

2.78M

hearing-impaired Indians

0

offline ISL translators

The reality

Every existing ISL translation app is **cloud-dependent**. Camera frames and mic audio leave the device. Latency is 1–3 seconds. The moment Wi-Fi drops — in a hospital ward, a government office, a rural school — **the conversation stops**.

What exists today

- Cloud-only translation apps
- Stable internet required
- Camera + mic data uploaded
- Fails in critical services

The gap is not capability. It is execution.

Why Cloud Translation Fails the People Who Need It Most

Requirement	Cloud App	On-Device	SignSync
Works without internet	**No**	Partial	**Yes**
Data leaves device	**Always**	Sometimes	**Never**
Latency	1–3 s	200–800 ms	**$<$ 400 ms**
Bidirectional ISL	Rare	No	**Yes**
Runs on phone NPU	No	Partial	**Fully**
Usable in hospitals	**No**	Maybe	**Yes**

Existing tools optimize for the cloud. **SignSync optimizes for the user.**

SignSync: Two Pipelines, One Phone

ISL → Text / Speech

Camera → MediaPipe → LSTM → Gemma 2B → Piper TTS

- Hand + body landmarks in real time
- Quantized classifier on NPU
- Gemma 2B refines grammar
- Piper TTS speaks the sentence

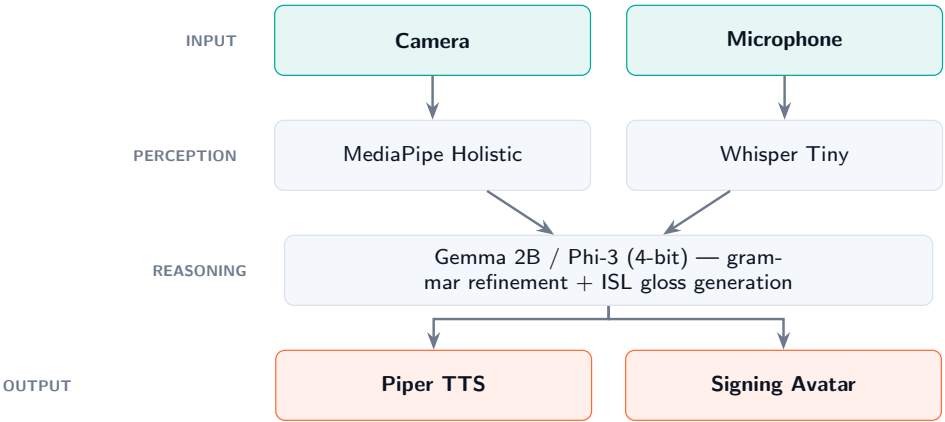
Speech → ISL

Mic → Whisper → Gemma 2B → Avatar

- Whisper Tiny transcribes offline
- LLM converts text → ISL glosses
- Animated avatar signs the reply
- Full loop closes on the device

Every model runs on the Snapdragon NPU. Nothing leaves the phone.

System Architecture



5 models. 1 phone. 0 bytes uploaded.

The On-Device AI Stack

Layer	Model / Tool	Why this choice
Vision	MediaPipe Holistic	Real-time landmarks, no GPU needed
Classification	Quantized LSTM (TFLite / ONNX)	Small, accurate on ISL sequences
Speech in	Whisper Tiny	Offline STT, robust to Indian accents
Speech out	Piper TTS	Natural voice, fully offline
Language	Gemma 2B / Phi-3, 4-bit	Grammar + ISL gloss generation
Runtime	ONNX Runtime + NNAPI	Snapdragon NPU, no cloud calls

4-bit quantization shrinks Gemma 2B from $\sim$2 GB to $\sim$1 GB — fits the phone, runs on the NPU.

Modular cascade. Ship vision $\rightarrow$ text on Day 1, add speech + avatar without rewriting the pipeline.

Why On-Device Is the Whole Point

1

Privacy

Camera and mic data never leave the phone. A conversation in a hospital room stays in that room.

2

Reliability

No internet, no problem. Works in rural schools, government offices, and disaster zones.

3

Latency

No round trip to a server. Sub-400 ms responses make conversation feel natural.

Cloud AI optimizes for scale. SignSync optimizes for the person in front of you.

Alignment

Criterion	How SignSync delivers
Phone-First	Core loop is camera + mic + NPU. No laptop, no server, no Wi-Fi.
AI-Native Build	A cascade of five on-device models — not a wrapper around a cloud API.
Office Kit	Screen mirroring for live model debugging; clipboard sync for test datasets.
Real Impact	Immediate utility for 2.78M people in hospitals, schools, and public offices.
Technical Depth	Quantization, NPU delegation, MediaPipe, LSTM, Whisper, Gemma — all on-device.

30-Hour Build Plan

Hours 0–8 • Vision

- MediaPipe + LSTM pipeline
- First end-to-end demo

Hours 8–16 • Language

- Gemma 2B on NPU
- 4-bit quantization

Hours 16–24 • Speech

- Whisper Tiny + Piper TTS
- Avatar signing loop

Hours 24–30 • Polish

- Latency tuning
- Live demo rehearsal

Every stage produces a demoable artifact.

Team sazi

Alden Antony

Lead & On-Device ML Engineer

- Android (Kotlin / Compose)
- On-device ML: TFLite, MediaPipe
- LLM quantization: llama.cpp, ONNX
- NPU delegation on Snapdragon

Why this team, this idea

- We have shipped on-device ML before this hackathon.
- The architecture is modular — every layer is testable alone.
- 30 hours is enough because the plan is not aspirational.
- This is not a demo. It is the first ISL translator that respects the user's privacy.

We are building a future where communication has no barriers.

Thank You

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